ESG-451 Weekly Status Update
PhytoPi

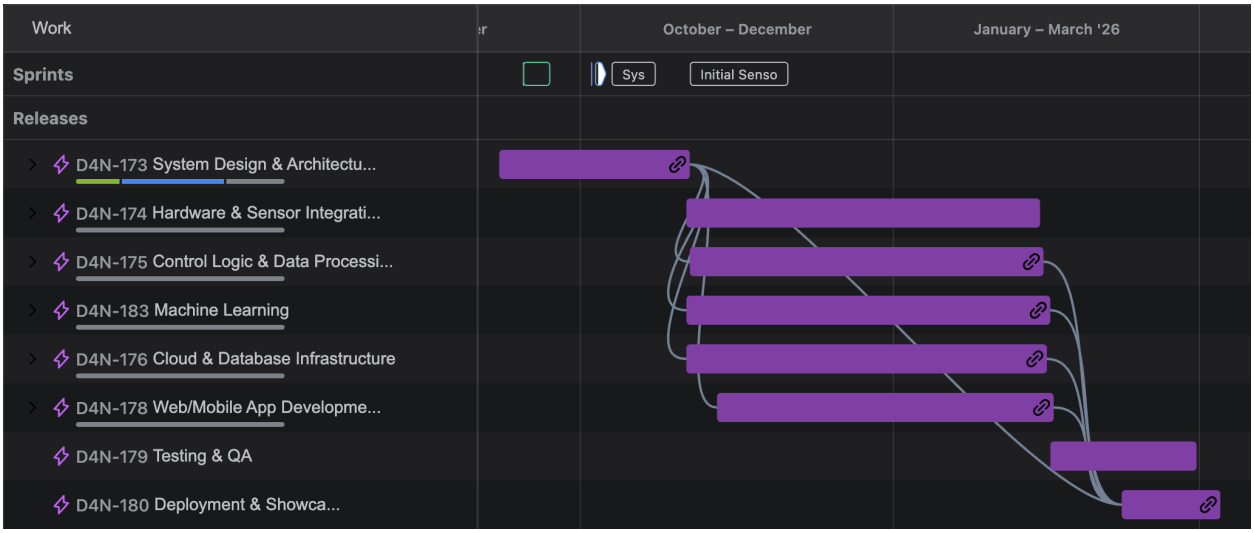
1. Project Description

PhytoPi is an IoT-based controlled environment system for supporting a plant throughout its entire growth cycle with minimal human intervention. Utilizing embedded hardware and a machine learning system monitoring the plant using a camera, PhytoPi is designed to automatically care for more than 80% of a plant's required maintenance over its lifespan.

2. Project Status Summary — Percent Complete 15%

The PhytoPi project is on-track for completion on the projected timeline. Currently, the Hardware/Integration Lead is putting together the project's portfolio and beginning development of temperature and humidity collection and timestamping software. The Software/Cloud Lead is currently completing database diagrams including entity relationship diagrams and database schema creation for each type of data collected such as temperature or light intensity.

3. Timeline Snapshot



4. Current Concept

The current design concept (see Figure 1) includes an improved design from version 0.2 by enforcing N-layer architecture design by separating out the system into 4 different layers. This enhances data security by making sure the business logic layer alone handles database access for sensor control. In Figure 2 we created a prototype of our physical product architecture.

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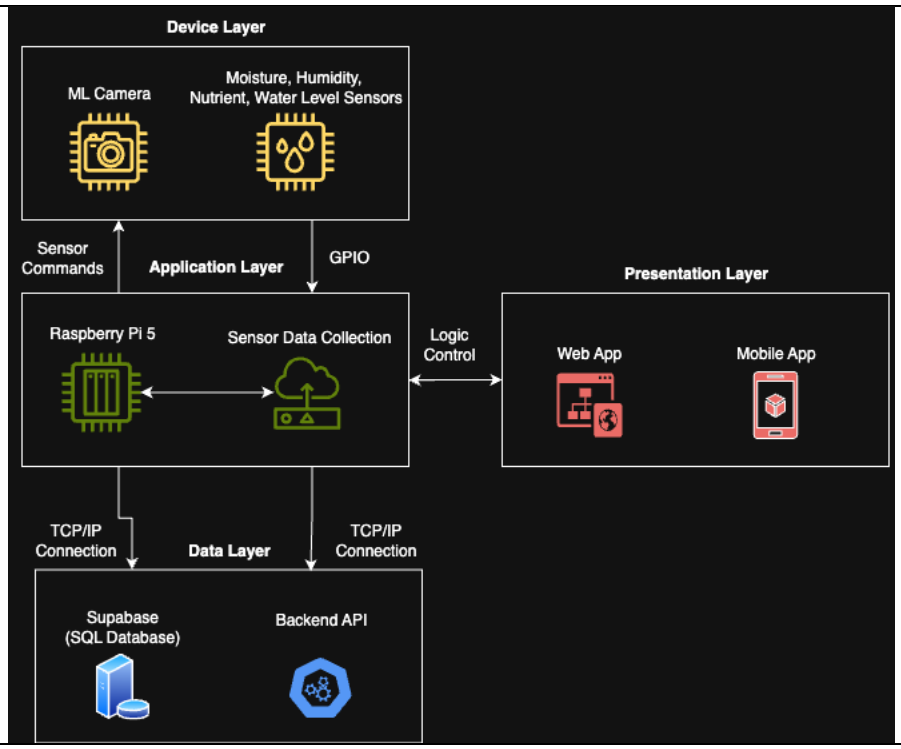

Figure 1. Current Design Concept - System Architecture Version 0.2

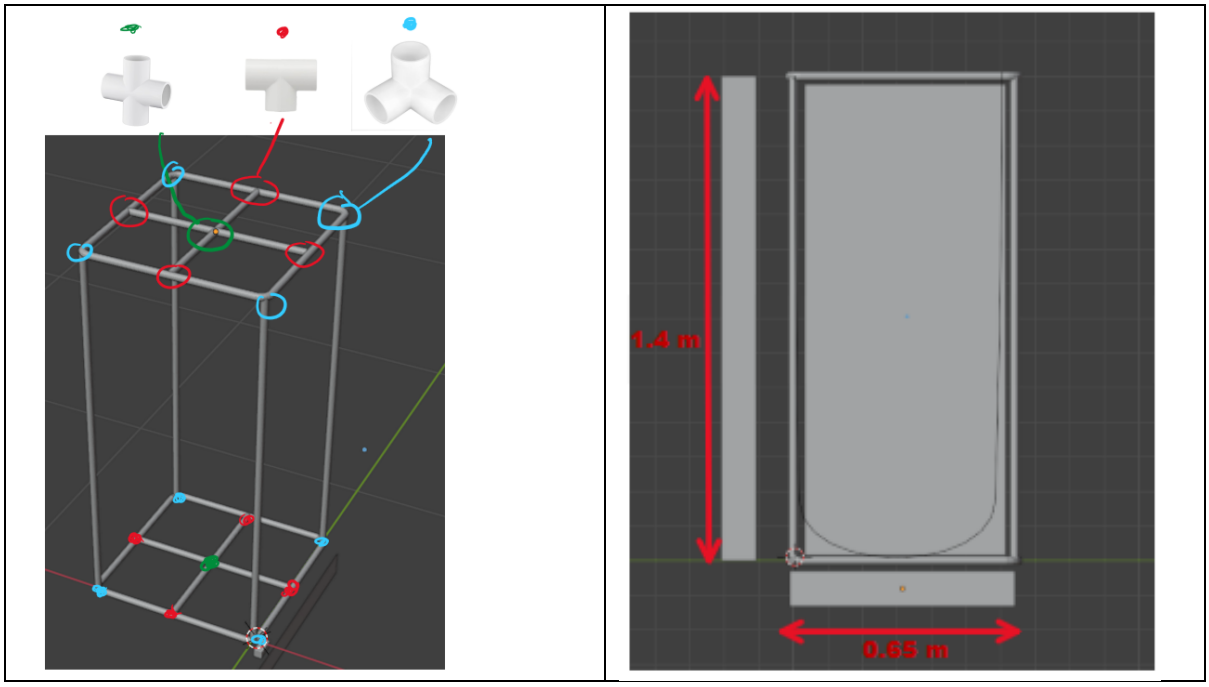

Figure 2. Current Design Concept – 3D Model of Final Product Version 0.1

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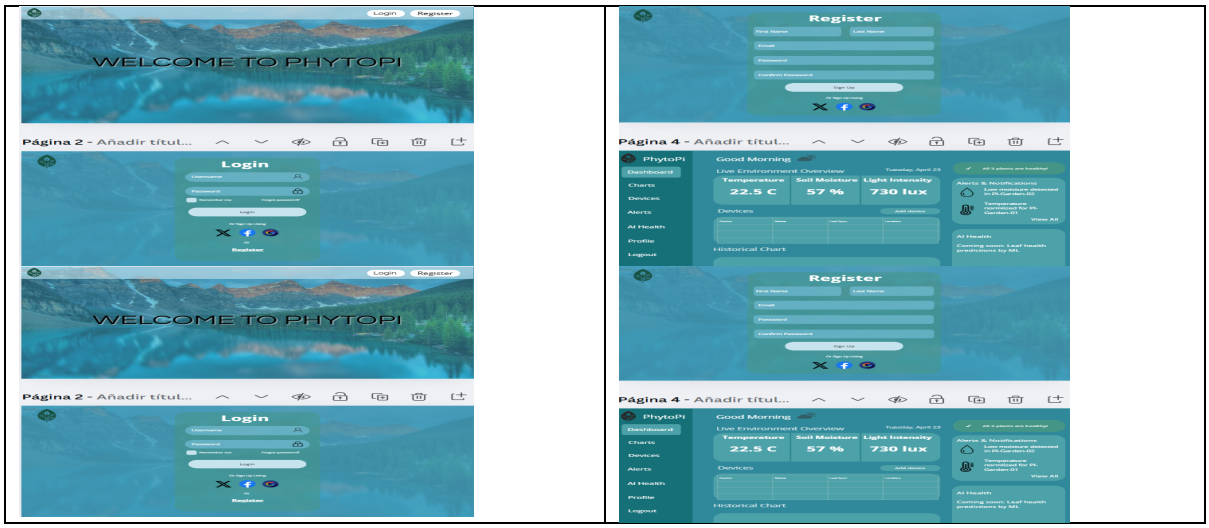

Figure 3. Current Design Concept – User Interface Version 0.1

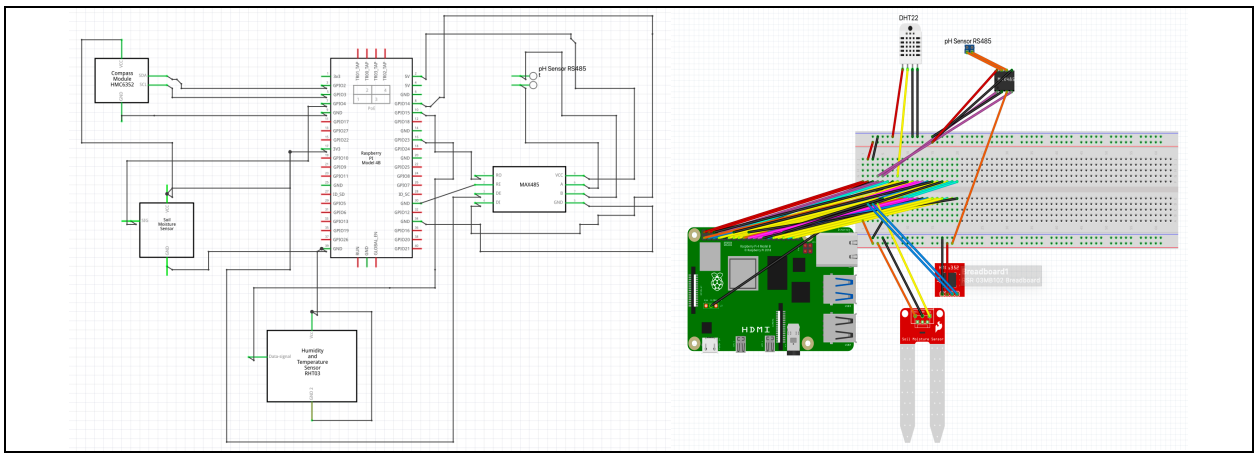

Figure 4. Current Design Concept – Wiring Diagram v0.3

5. Accomplishments and Learning

Planned from last week:

- Electrical design – completed
- Figma UI design – completed
- Finalize all project requirements – 80% complete
- Privacy agreement write-up – 10% complete
- Entity Relationship Diagram – 50% complete
- Database Schema – 50% complete
- Temperature/Humidity Collection Logic – 15% complete
- Project Portfolio – 25% complete

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Unplanned:

- Light Intensity Collection Logic
- Soil Moisture Collection Logic
- Plant Pot CAD Design
- Send Sample Readings to Database

Learning:

- Fritzing can be used as a detailed design software for our entire hardware layout including sensors and the Raspberry Pi 5.
- Supabase has many tools available not only for setting row-level rules for data storage but it also for automated schemas and entity relationship diagrams.
- Coordinating team meetings is difficult outside of class time, best time for team meetings is typically immediately after class or over video call.
- Cloud-based document storage on OneDrive is crucial for concurrent progress on crucial documents such as our System Design Presentation and weekly updates.

6. Work Planned for Next Week

- Critical Path – Finish the project portfolio and begin rapid prototyping of 3D printing plant pot enclosures with integrated sensor slots.
- Finish the database design completely, which involves an entity relationship diagram and schemas for each type of data collected.
- Test and calibrate all available sensors for accurate data collection/compatibility.
- Test timestamping, recording, and moving collected sensor data efficiently in a format acceptable for moving to UI display.
- Set row-level rules for temperature and humidity values in database.
- Complete privacy agreement write up.

7. Risks

- CAD design for plant pot might prove to be more difficult than expected with the lack of 3D modeling experience between both team members. This can be mitigated by making rapid prototyping for this a critical path for the project and emphasizing the importance of effort into this design.
- Without careful consideration of the entity relationships within the database, it may be the case that the database structure does not function as intended for data storage or does not follow the intended structure of our tables. This would require refactoring of the database structure, resulting in a time delay of up to one week.

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8. Deliverables and Milestones

Milestone	Planned	Forecasted	Actual	Status
Database Infrastructure	11/30/25	11/30/25	N/A	On Schedule
Complete System Design	10/31/25	11/1/25	N/A	On Schedule
Complete Initial Sensor Testing	11/30/25	11/30/25	N/A	On Schedule
Deliverable	Planned	Forecasted	Actual	Status
Project Portfolio	10/31/25	10/31/25	N/A	On Schedule
Technical Specifications	11/2/25	11/2/25	N/A	On Schedule
Database Entity Relationship Diagram	11/1/25	11/1/25	N/A	On Schedule
Timestamped Sample Readings for Temperature	11/12/25	11/12/25	N/A	On Schedule

9. Attachments

N/A

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Approvals:

Team Role	Print Name	Signature	Date
Product Manager/Hardware + Integration Lead	Nolan Tuttle	Nolan Tuttle	10/31/25
Software + Cloud Lead	Daniel Grijalva	Daniel Grijalva	10/31/25
Faculty Mentor	Bill Hughes	Bill Hughes	10/31/25

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